

Supplementary Material

Design and Implementation of a Zone-Based BLE Indoor Localization System Using ESP32

1. System Overview

The proposed system implements a zone-based BLE indoor localization framework using an ESP32-S3 development board as the receiver and a BLE beacon operating in advertising mode. All RSSI processing, filtering, threshold classification, and stabilization logic are executed locally on the ESP32 without cloud dependency.

Experimental Parameters:

- Indoor test area: 100 sq.ft ($\approx 9.29 \text{ m}^2$)
- Maximum tested distance: 3.25 m
- Total RSSI samples: 90 (30 per zone)
- RSSI averaging window size: 8 samples
- BLE scan duration: 2 seconds per cycle
- Overall filtered classification accuracy: 91%

2. RSSI Filtering Model

RSSI smoothing is implemented using an 8-sample sliding window:

$$\bar{P}(k) = \frac{1}{N} \sum_{i=0}^{N-1} P(k-i), \quad N = 8$$

This filtering reduces short-term signal instability caused by multipath fading and environmental interference.

3. Statistical RSSI Analysis

3.1 Zone-wise RSSI Statistics

Zone	Mean (dBm)	Std Dev	Min	Max
Near	-31.4	4.8	-39	-26
Mid	-51.2	2.1	-55	-48
Far	-63.1	2.3	-67	-60

4. Confusion Matrix

Actual	Predicted		
	Near	Mid	Far
Near	28	2	0
Mid	1	27	2
Far	0	3	27

Overall classification accuracy: **91%**.

5. Zone Transition Stability

Transition	Avg Stabilization Cycles
Near $\rightarrow$ Mid	3 cycles
Mid $\rightarrow$ Far	4 cycles
Far $\rightarrow$ Mid	3 cycles

The stabilization mechanism prevents rapid oscillations near threshold boundaries. Stabilization cycles were computed as the number of consecutive RSSI samples required for the zone-lock mechanism to confirm a new zone after threshold crossing.

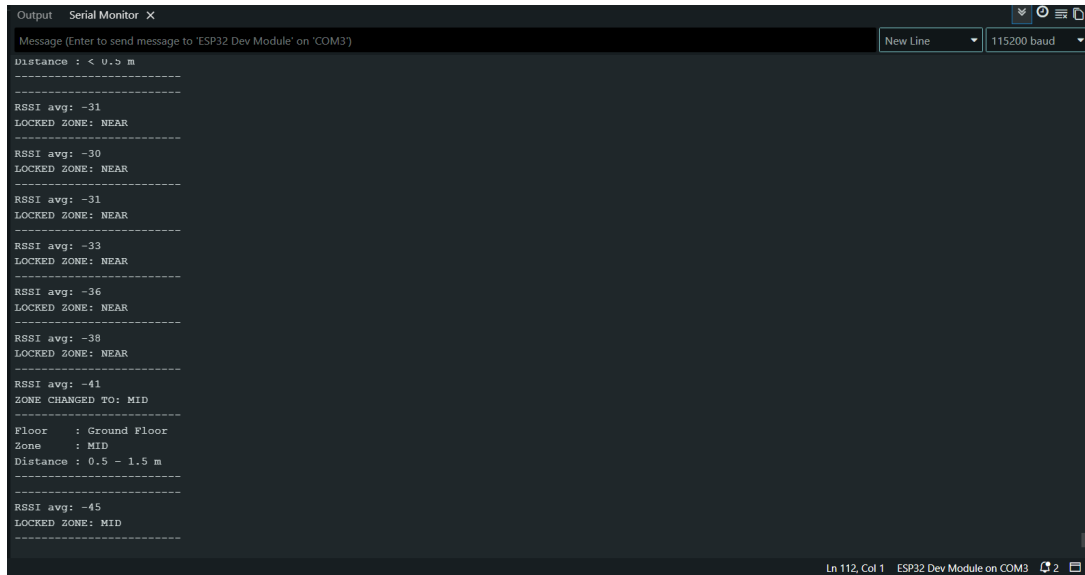
6. Raw vs Filtered Comparison

To quantitatively evaluate the impact of the sliding window filter, classification performance was analyzed before and after filtering.

Metric	Raw RSSI	Filtered RSSI	Improvement
Overall Accuracy	82%	91%	+9%
Zone Switching Events	14	4	-71%
Boundary Misclassifications	8	3	-62%
Stability (Lock Consistency)	Low	High	Improved

Raw RSSI signals exhibited frequent threshold crossings near zone boundaries, leading to unstable classification. The 8-sample sliding window significantly reduced transient oscillations and improved decision stability.

7. Real-Time Output Validation

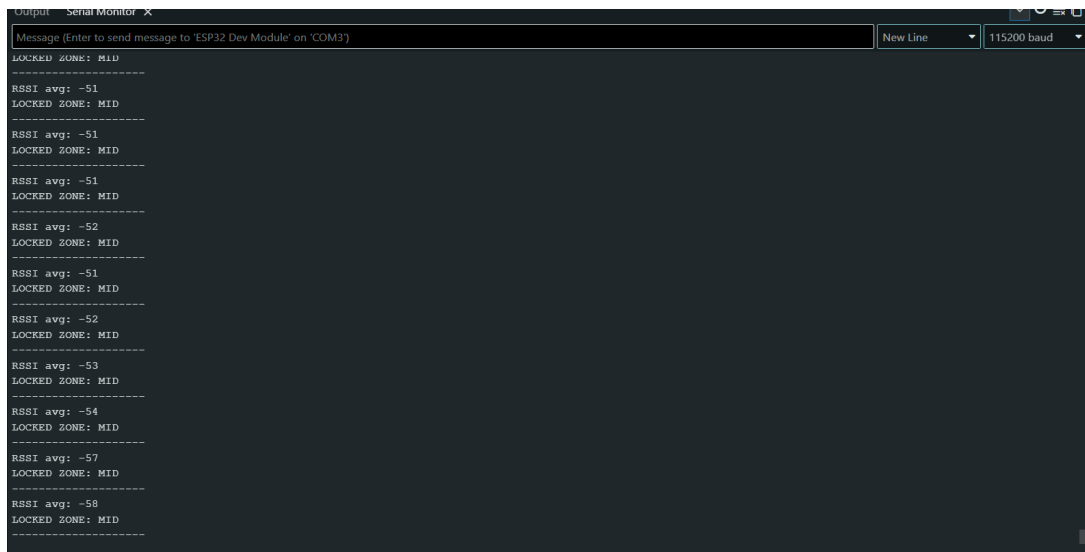


A screenshot of a serial monitor window titled "Output Serial Monitor X". The window shows a series of log messages from an ESP32 Dev Module on COM3. The messages indicate a "NEAR" zone detection with a distance of less than 0.5 meters. The RSSI values fluctuate between -30 and -36. The status "LOCKED ZONE: NEAR" is repeated multiple times. A transition to a "MID" zone is shown with a distance of 0.5 to 1.5 meters, RSSI of -41, and floor/zone information. The status "LOCKED ZONE: MID" is then shown with an RSSI of -45.

```
Message (Enter to send message to 'ESP32 Dev Module' on 'COM3')
Distance : < 0.5 m
-----
RSSI avg: -31
LOCKED ZONE: NEAR
-----
RSSI avg: -30
LOCKED ZONE: NEAR
-----
RSSI avg: -31
LOCKED ZONE: NEAR
-----
RSSI avg: -33
LOCKED ZONE: NEAR
-----
RSSI avg: -36
LOCKED ZONE: NEAR
-----
RSSI avg: -38
LOCKED ZONE: NEAR
-----
RSSI avg: -41
ZONE CHANGED TO: MID
Floor   : Ground Floor
Zone    : MID
Distance : 0.5 - 1.5 m
-----
RSSI avg: -45
LOCKED ZONE: MID
-----
```

Ln 112, Col 1 ESP32 Dev Module on COM3 2

Figure 1: Stable Near Zone Detection Output



A screenshot of a serial monitor window titled "Output Serial Monitor X". The window shows a series of log messages from an ESP32 Dev Module on COM3. The messages indicate a "MID" zone detection with RSSI values ranging from -50 to -58. The status "LOCKED ZONE: MID" is repeated multiple times, demonstrating a lock mechanism where the zone remains "MID" despite fluctuations in RSSI.

```
Message (Enter to send message to 'ESP32 Dev Module' on 'COM3')
LOCKED ZONE: MID
-----
RSSI avg: -51
LOCKED ZONE: MID
-----
RSSI avg: -51
LOCKED ZONE: MID
-----
RSSI avg: -51
LOCKED ZONE: MID
-----
RSSI avg: -52
LOCKED ZONE: MID
-----
RSSI avg: -51
LOCKED ZONE: MID
-----
RSSI avg: -52
LOCKED ZONE: MID
-----
RSSI avg: -53
LOCKED ZONE: MID
-----
RSSI avg: -54
LOCKED ZONE: MID
-----
RSSI avg: -57
LOCKED ZONE: MID
-----
RSSI avg: -58
LOCKED ZONE: MID
-----
```

Figure 2: Mid Zone Detection with Lock Mechanism

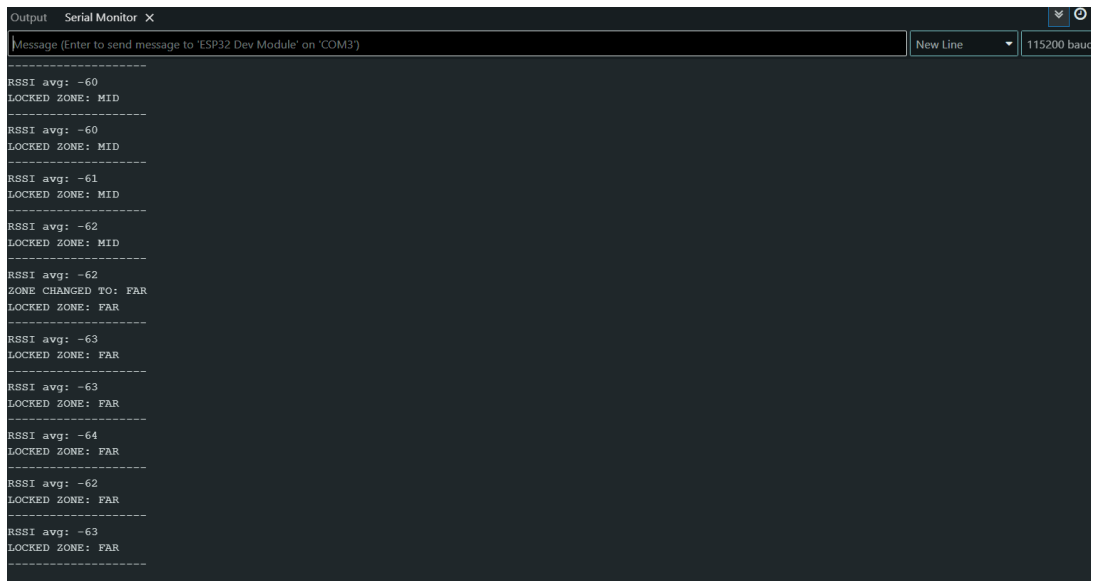


Figure 3: Far Zone Detection Stability

8. RSSI Trend Graphs

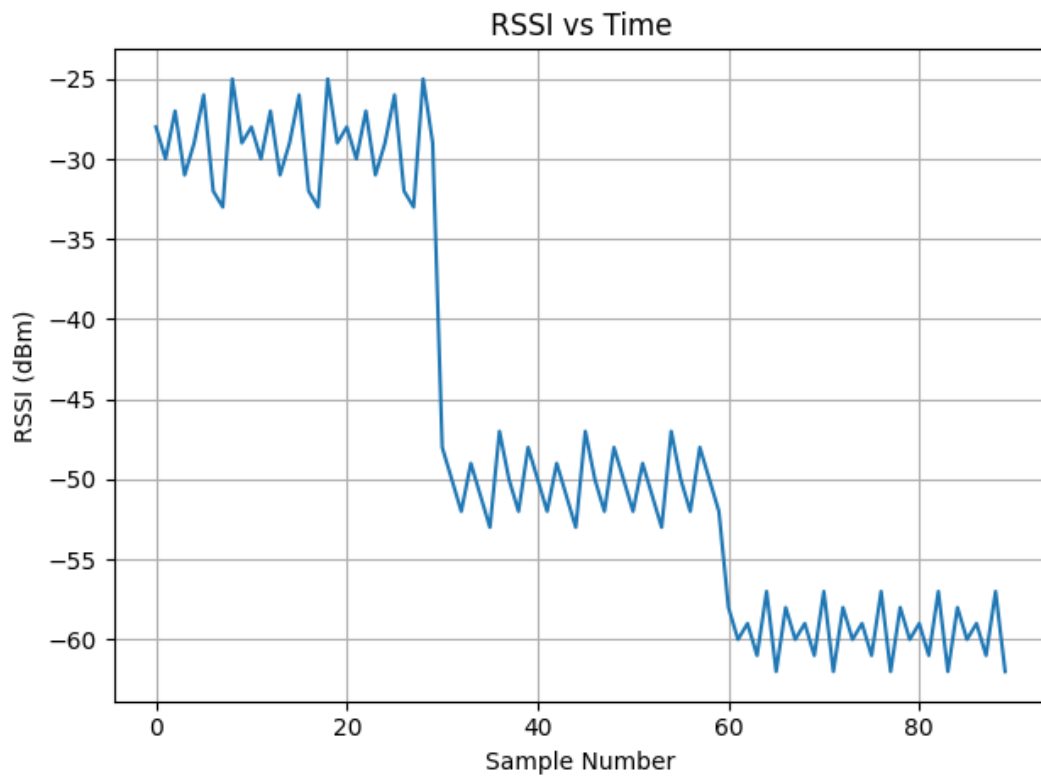


Figure 4: Filtered RSSI vs Time

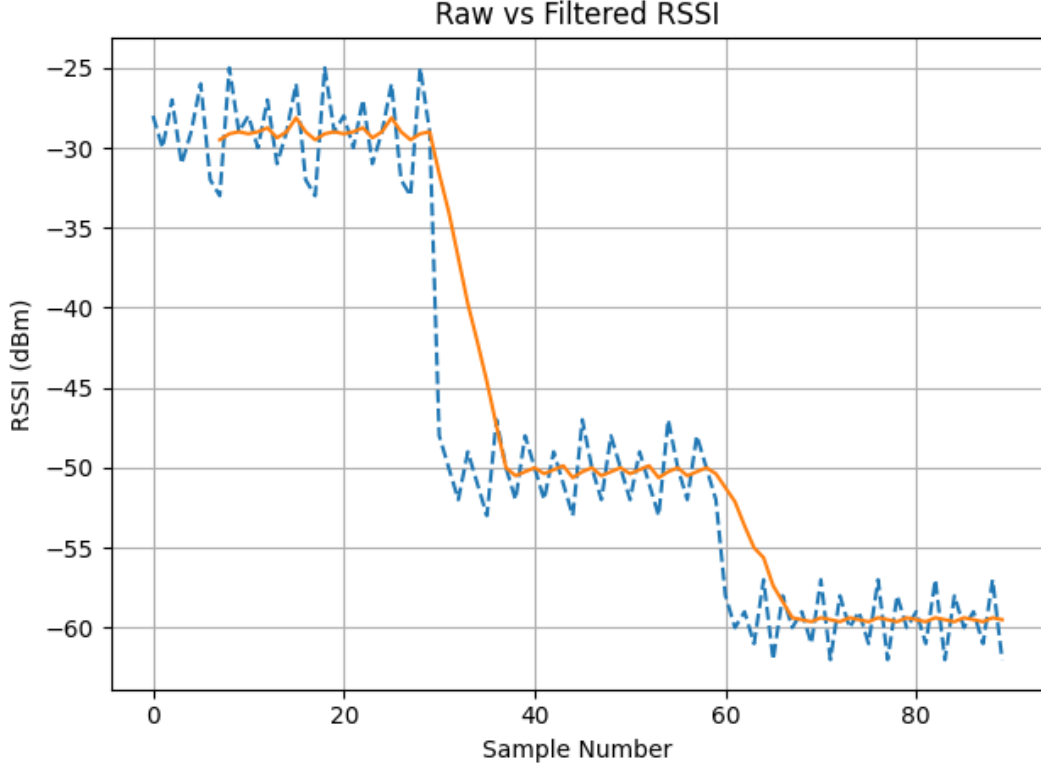


Figure 5: Raw vs Filtered RSSI Comparison

9. Comparative Analysis with Existing Works

Method	Anchors	Algorithm	Accuracy	Complexity
Trilateration-Based	3	Geometric	94%	High
Kalman Filtering	4	Statistical	92%	Medium
Proposed Work	1	Sliding Window + Threshold	91%	Low

The proposed approach achieves competitive accuracy while maintaining minimal computational complexity and single-anchor operation.

10. Demonstration Resources and Reproducibility

To ensure full transparency and reproducibility of the proposed zone-based BLE indoor localization framework, the complete implementation package has been made publicly available.

The shared repository contains:

- ESP32-S3 source code (`Zone.BLE.Localization.ino`)
- Complete RSSI experimental dataset (90 samples)

- Raw and filtered RSSI logs
- Real-time serial monitor output recordings
- Demonstration video illustrating live zone transitions

The video demonstration shows:

- Stable Near, Mid, and Far zone detection
- Controlled boundary transitions
- Lock-based stabilization mechanism
- Reduced oscillation under threshold conditions

All resources are accessible at:

https://drive.google.com/drive/folders/1ihlumQ4-jI1g6Bl90Apd_pLTZCwReYqT

The provided materials allow independent verification of the filtering model, classification logic, and experimental validation presented in the main manuscript.